

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 24/05/2018

Day: Thursday

Time: 10:00 am to 10:15 am

Total Marks: 10

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : BMRIT/BOTAT- May 2018
Semester: 2
Section : I
Course Code : BMRIT/BOTAT 202
Course Name: Basic physics Including Radiological Physics

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs)

(10 × 1 = 10)

(Encircle the correct answer)

1. Force applied is 15 Newton and displacement is about 25 meters. Work done is?
 - a. 200 Joule
 - b. 375 Joule
 - c. 200 Pound
 - d. 375 Pound
2. Atom always has electrically Charged.
 - a. Positive
 - b. Negative
 - c. Neutral
 - d. None of the above
3. The energy required to remove one electron from a single electron is called.....
 - a. Kinetic energy
 - b. Potential energy
 - c. Work function
 - d. Ionization energy

4. **Mass energy equivalence is the famous relation given by.....**
- $E^2 = mc$
 - $E^2 = mc^2$
 - $E = mc^2$
 - $E = m^2c$
5. **Electronic component that resist the electric current flowing through it is called.....**
- Inductor
 - Resistance
 - Transistor
 - Capacitor
6. **Kirchhoff's junction rule is the statement of the conservation.**
- Energy
 - Voltage
 - Power
 - Charge
7. **If two waves collide and if their amplitudes are subtracted it is called**
- Destructive Interference
 - Constructive interference
 - Coherence
 - Diffraction
8. **In diagnostic ultrasound, dark grey image on the grey-scale image is called,**
- Hyperechoic
 - Hypoechoic
 - Anechoic
 - Enechoic
9. **At absolute zero degree temperature a semiconductor behaves like an**
- Metal
 - Semimetal
 - Superconductor
 - Insulator
10. **In n-type semiconductor.....are the majority charge carriers in the n-type semiconductor.**
- Electrons
 - Free electrons
 - Holes
 - None of the above

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B.M.R.I.T/B.O.T.A.T Examination

May 2018

BMRIT/BOTAT 202 Basic Physics Including Radiological Physics

Date: 24/05/2018

Time: 10:15 am to 12:00 pm

Maximum Marks: 25

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 6) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Define Work and Power (with equation and SI units).
- b. Define excitation energy and binding energy.
- c. State Kirchhoff loop rule.
- d. Write four properties of ultrasonic wave.
- e. Define heat capacity and specific heat capacity.
- f. Give two differences between intrinsic and extrinsic semiconductor.
- g. Define p-type and n-type semiconductor.

Q. 2. Short Notes (3 out of 5) (Answer not exceeding 150 words) (3 × 5 = 15)

- a. Explain Ionization, excitation and photoelectric effect.
- b. Define Alternating and direct current. Explain the working of AC generator.
- c. Explain Maxwell equations for electromagnetic wave.
- d. Explain Doppler's effect.
- e. Explain three phenomenon (technique) for heat transfer.